

ECA5601 –Advanced Wireless Communication Systems / 5G / 6G

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Experiment: 10

5G Network Slicing Environment Using MATLAB

DATE:12-8-26

Aim:To analyze the performance of a 5G Network Slicing environment by studying the number of network slices, slice utilization, and end-to-end latency using MATLAB.

Objective : Analyze the Number of Network Slices

Analyze the creation and allocation of multiple network slices in a 5G network using MATLAB.

Observe how different slices can be configured to support diverse service requirements and applications.

Software Required : • MATLAB

Theory:

Network slicing is a key feature of 5G that enables a single physical network infrastructure to be partitioned into multiple logical networks called slices. Each slice is optimized for a specific service category, such as enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC), or massive Machine-Type Communications (mMTC). By dynamically allocating network resources, network slicing improves flexibility, resource utilization, and Quality of Service (QoS) while supporting diverse application requirements.

MATLAB Code:

```
clc;
clear;
close all;

slices = {'eMBB','URLLC','mMTC','Private','IoT'};

count = [1 1 1 1 1]; % One instance of each network slice

figure
bar(count)
```

grid on

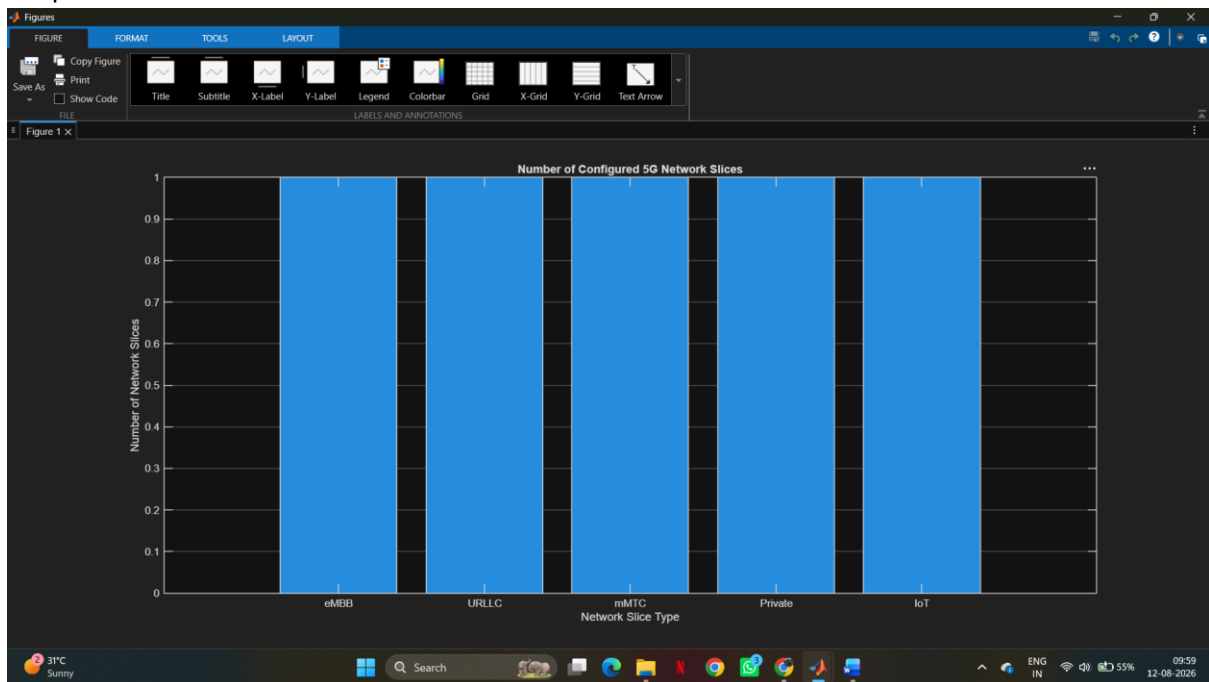
```
set(gca,'XTickLabel',slices)
```

```
xlabel('Network Slice Type')
```

```
ylabel('Number of Network Slices')
```

```
title('Number of Configured 5G Network Slices')
```

Output:



Result:

The MATLAB analysis successfully demonstrated the characteristics of a 5G Network Slicing environment. The results showed that multiple network slices can coexist while exhibiting different utilization levels and latency characteristics based on their intended services.